

Artificial Intelligence as a General-Purpose Technology

Institutional Enablement Across Economic, Political, and Epistemic Systems

Soumadeep Ghosh and GPT-5.5 Thinking

Kolkata, India

Abstract

Artificial intelligence is often discussed as if it were an autonomous mind, a substitute institution, or a rival source of authority. This paper offers a different interpretation. Artificial intelligence is best understood as a general-purpose technology that transforms the way individuals interact with computers. Its central novelty is not that computers acquire human-like thought, but that computational systems become capable of processing representations of human thought through natural language. AI therefore enables three connected transformations: natural-language computing, represented-thought processing, and institutional knowledge acceleration. These transformations affect firms, universities, courts, bureaucracies, laboratories, media organizations, and political institutions by lowering the cost of querying, organizing, summarizing, analyzing, auditing, and generating documents and decisions. Yet institutional adoption is constrained by philosophical concerns over agency, authorship, authenticity, accountability, judgment, and knowledge. The paper argues that such concerns are serious but not sovereign. Philosophy itself is institutionally situated and can become self-defeating when it exempts itself from the standards it applies to AI. At a deeper level, philosophy may become its own anathema when it reproduces the dogmatism, opacity, exclusion, and authority it claims to critique. The correct institutional response is neither blind adoption nor philosophical rejection, but accountable integration under reciprocal scrutiny.

The paper ends with “The End”

Contents

1	Introduction	2
2	Artificial Intelligence as a General-Purpose Technology	3
3	Natural-Language Computing	3
4	Represented-Thought Processing	4
5	Institutional Knowledge Acceleration	4
6	A Simple Model of AI-Enabled Institutions	5
7	The Philosophical Hesitation	5
7.1	Agency	6
7.2	Authorship	6
7.3	Authenticity	6
7.4	Accountability	6
7.5	Knowledge	6
8	The Self-Defeat of Philosophical Objection	6
9	Philosophy as Its Own Anathema	7
10	Accountable Integration	7
10.1	Algorithm for Institutional AI Use	7

11 A Diagram of the Argument	8
12 Implications	8
13 Conclusion	8

List of Figures

1 AI as a general-purpose technology producing institutional enablement and philosophical contestation.	8
---	---

List of Tables

1 AI functions across institutional settings	5
--	---

1 Introduction

Artificial intelligence has entered institutional life through a peculiar doorway. It is not merely a new machine, nor merely a new software layer, nor merely a statistical instrument. It changes the relationship between human beings and computers. Earlier computing systems required users to translate their intentions into formal commands, menus, database queries, programming languages, or rigid workflows. AI systems based on natural language weaken that barrier. A person can ask, revise, compare, summarize, draft, classify, search, and reason with a machine in ordinary language.

This does not imply that the machine thinks as a human being thinks. A computer has no human biography, no lived interiority, no organic consciousness, and no moral experience of its own. Yet it can store, retrieve, transform, and generate representations of human thought. These representations include books, papers, contracts, court judgments, code, emails, laws, minutes, transcripts, reports, datasets, and institutional records. AI becomes institutionally significant because institutions are already built out of represented thought. A university is not only a campus; it is also syllabi, lectures, manuscripts, examinations, committees, citations, curricula, and archives. A court is not only a judge; it is pleadings, statutes, precedents, procedures, evidentiary records, and judgments. A government is not only officials; it is memoranda, rules, forms, budgets, reports, orders, and files.

The thesis of this paper is therefore as follows. Artificial intelligence is a general-purpose technology that transforms human-computer interaction through natural-language computing, represented-thought processing, and institutional knowledge acceleration. Its institutional importance lies not in being an institution itself, but in enabling institutions to access, reorganize, audit, and generate their own represented thought at lower cost.

However, the adoption of AI is not purely a technical matter. Institutions often hesitate because AI touches philosophical questions: Who is the author of an AI-assisted text? Who is responsible for an AI-assisted decision? Does AI generate knowledge or only plausible language? Does its use diminish human authenticity? Should institutions allow machine-generated drafts, summaries, or judgments into domains where legitimacy depends on human responsibility?

These objections are important. Yet they are not final. Philosophy itself is not outside institutional life. Philosophical objections are produced by persons, departments, journals, traditions, committees, incentives, professional hierarchies, and intellectual histories. If philosophy demands that AI be audited for opacity, bias, power, and legitimacy, then philosophy must accept equivalent scrutiny of its own procedures. When philosophy refuses that reciprocity, it becomes self-defeating. When it becomes the dogma, opacity, exclusion, or authority it condemns, it becomes its own anathema.

The argument proceeds as follows. Section 2 defines artificial intelligence as a general-purpose technology. Section 3 explains natural-language computing. Section 4 develops the idea of represented-thought processing. Section 5 models institutional knowledge acceleration. Section 6 discusses philosophical hesitation. Section 7 develops the critique of philosophical self-defeat and philosophical anathema. Section 8 proposes a framework for accountable integration. Section 9 concludes.

2 Artificial Intelligence as a General-Purpose Technology

General-purpose technologies are technologies with broad applicability, scope for continuous improvement, and complementarities with many sectors. Steam power, electricity, the internal combustion engine, computing, and the internet are standard examples. They do not merely perform one task; they reorganize production, communication, administration, and coordination across fields.

Definition 1 (General-purpose technology). *A technology G is a general-purpose technology if it satisfies three conditions:*

1. **Pervasiveness:** *G can be applied across many sectors.*
2. **Improvement:** *the performance of G can improve over time.*
3. **Complementarity:** *the value of G increases when institutions reorganize their practices, skills, and capital around it.*

Artificial intelligence satisfies these conditions. It is pervasive because it can be applied in education, law, medicine, finance, government, science, media, software, logistics, and personal productivity. It improves because models, data, hardware, alignment methods, interfaces, and deployment architectures continue to change. It is complementary because AI becomes valuable only when embedded in workflows: document management, scientific discovery, customer service, programming, compliance, research, teaching, translation, and institutional analysis.

Yet AI differs from earlier general-purpose technologies in its interface. Electricity transformed energy. Engines transformed motion. Computers transformed calculation and symbolic storage. The internet transformed communication and distribution. AI transforms the usability of represented thought by allowing individuals to interact with computational systems through ordinary language.

Proposition 1 (AI as interface-shifting GPT). *Artificial intelligence is a general-purpose technology whose distinctive institutional contribution is the transformation of human-computer interaction from formal command to natural-language representation.*

Proof. A general-purpose technology affects many domains through complementary reorganization. AI does so by entering domains wherever symbolic material is important. Its distinctive interface effect is that users need not express all tasks in formal code, database syntax, or rigid menus. They can express intentions in ordinary language. Since most institutional knowledge is already stored in natural or semi-natural language, AI reduces the translation cost between human intention and computational operation. Therefore AI is a general-purpose technology whose distinctive contribution is interface transformation. $\square$

3 Natural-Language Computing

Traditional computing requires formalization. A spreadsheet requires formulas. A database requires queries. A programming language requires syntax. A bureaucracy requires forms. A search engine requires keywords. AI weakens the distance between human intention and machine operation.

Let I denote a human intention, C denote a computational operation, and τ denote the translation cost between them. In traditional computing, the user must often convert I into a formal representation F before C can occur:

$$I \longrightarrow F \longrightarrow C.$$

With natural-language AI, the user may express I in ordinary language L :

$$I \longrightarrow L \longrightarrow C.$$

The institutional significance lies in the reduction:

$$\tau(I, L) < \tau(I, F)$$

for many ordinary users and many document-centered tasks.

Definition 2 (Natural-language computing). *Natural-language computing is the use of ordinary language as an operational interface between human intention and computational execution.*

Natural-language computing does not eliminate formal systems. Rather, it allows users to invoke them indirectly. A lawyer may ask for a summary of a contract. A researcher may ask for a literature map. A bureaucrat may ask for inconsistencies across files. A student may ask for feedback on an essay. A programmer may ask for code generation. A manager may ask for a comparison of reports. The machine remains computational, but the interface becomes linguistic.

4 Represented-Thought Processing

The phrase “represented thought” refers to symbolic traces produced by human beings and institutions. It does not require that the machine possess thoughts. It only requires that human thought has been externalized into symbols.

Definition 3 (Represented thought). *Represented thought is human or institutional cognition externalized into symbolic form, including text, code, law, mathematics, data, diagrams, speech transcripts, archives, and procedural records.*

AI systems process represented thought in several ways:

1. **Querying:** locating relevant information inside large corpora.
2. **Organization:** grouping, classifying, labeling, and structuring documents.
3. **Understanding:** producing interpretations, explanations, and conceptual maps.
4. **Analysis:** identifying patterns, inconsistencies, arguments, assumptions, and implications.
5. **Summarization:** compressing long material into shorter representations.
6. **Generation:** producing new drafts, arguments, tables, models, code, plans, or hypotheses.

The institutional importance of AI follows from the fact that institutions are heavily composed of represented thought. A firm consists partly of contracts, accounts, memos, strategy documents, organizational charts, emails, databases, and standard operating procedures. A state consists partly of laws, regulations, administrative orders, intelligence reports, files, budgets, and statistical systems. A university consists partly of articles, syllabi, lectures, grades, theses, committee minutes, and citations.

Proposition 2 (Institutional symbolic dependence). *The more an institution depends on represented thought, the greater the potential productivity effect of AI-enabled represented-thought processing.*

Proof. Let institution j have output Y_j depending on physical capital K_j , labor L_j , and represented thought R_j :

$$Y_j = A_j F(K_j, L_j, R_j).$$

Suppose AI increases the effective usability of represented thought from R_j to ϕR_j , where $\phi > 1$. Then output becomes

$$Y'_j = A_j F(K_j, L_j, \phi R_j).$$

If F is increasing in R_j , then $Y'_j > Y_j$. The magnitude of the gain depends on the marginal product of represented thought, $\partial F / \partial R_j$. Therefore institutions more dependent on represented thought gain more from AI-enabled processing. $\square$

5 Institutional Knowledge Acceleration

AI accelerates institutional knowledge by lowering the cost of operations performed on represented thought. Let D denote an institutional document corpus. Let the institution perform tasks

$$T = \{q, o, u, a, s, g\},$$

where q is querying, o is organizing, u is understanding, a is analyzing, s is summarizing, and g is generating.

Let $c_i(D)$ be the cost of performing task i on corpus D without AI, and let $c_i^A(D)$ be the cost with AI. The institutional acceleration condition is:

$$c_i^A(D) < c_i(D)$$

for a sufficiently large subset of tasks $i \in T$.

Definition 4 (Institutional knowledge acceleration). *Institutional knowledge acceleration occurs when AI lowers the time, cost, or cognitive burden of performing operations on institutional represented thought.*

The effect is not simply speed. It changes what institutions can practically know about themselves. A government ministry with millions of pages of documents may formally possess information that it cannot practically retrieve. A university may possess thousands of papers but lack a usable conceptual map. A court may possess precedents that are legally available but cognitively expensive to compare. A firm may possess internal reports that no manager has time to synthesize. AI changes the practical boundary between stored information and usable knowledge.

Table 1: AI functions across institutional settings

Institution	Corpus	AI operation	Institutional effect
University	Papers, syllabi, theses	Summarize, compare, map	Research acceleration
Court	Statutes, precedents, filings	Retrieve, classify, analyze	Legal consistency
Firm	Reports, contracts, emails	Organize, draft, audit	Managerial coordination
Government	Rules, files, budgets	Query, synthesize, monitor	Administrative capacity
Laboratory	Data, notes, articles	Hypothesize, code, explain	Discovery support
Media	Archives, transcripts, drafts	Summarize, generate, verify	Editorial productivity

6 A Simple Model of AI-Enabled Institutions

Consider an institution that produces output Y using labor L , capital K , and institutional knowledge M . Institutional knowledge is produced from represented thought R and processing capacity P :

$$M = H(R, P).$$

Without AI, processing capacity is P_0 . With AI, processing capacity becomes

$$P_A = P_0 + \alpha A,$$

where A denotes AI capability and α measures institutional integration.

Institutional output is:

$$Y = F(K, L, M) = F(K, L, H(R, P_0 + \alpha A)).$$

The key variable is not AI capability alone but integration. A powerful AI system with $\alpha = 0$ has no institutional effect. Conversely, a moderate AI system with high integration may generate large gains.

Proposition 3 (Integration complementarity). *AI capability increases institutional output only through institutional integration.*

Proof. Differentiate output with respect to AI capability:

$$\frac{\partial Y}{\partial A} = F_M(K, L, M)H_P(R, P_0 + \alpha A)\alpha.$$

If $\alpha = 0$, then $\partial Y/\partial A = 0$. If $\alpha > 0$, and both F_M and H_P are positive, then $\partial Y/\partial A > 0$. Hence AI capability produces institutional output gains only when integrated into institutional processes. $\square$

This proposition explains why AI adoption differs across institutions. Some institutions possess rich corpora but weak integration. Others possess strong integration but poor data governance. Others resist integration for philosophical, legal, ethical, or reputational reasons.

7 The Philosophical Hesitation

Institutions do not hesitate merely because AI is technically immature. They hesitate because AI unsettles inherited categories.

7.1 Agency

When AI produces a recommendation, who acts? The user? The model? The institution? The vendor? The developer? Institutional action usually requires identifiable responsibility. AI distributes agency across human and technical systems.

7.2 Authorship

If a scholar uses AI to draft a paragraph, is the scholar the author? If a lawyer uses AI to summarize precedent, is the legal argument still the lawyer's? If a bureaucrat uses AI to draft a memo, is the memo institutionally authored? AI complicates the boundary between assistance and authorship.

7.3 Authenticity

Some institutions value not only outcomes but the visible performance of human effort. A student essay, judicial opinion, philosophical argument, or literary work may be judged partly by the presumed human process behind it. AI assistance raises the question of whether the product remains authentic.

7.4 Accountability

AI can generate errors, hallucinations, biases, or unsupported claims. Institutions require mechanisms of responsibility. If AI output enters institutional decisions without audit, accountability becomes fragile.

7.5 Knowledge

Does AI know, or does it merely generate plausible symbolic sequences? This question matters in education, law, science, and public administration. A machine may produce a correct answer without possessing human understanding. Institutions must decide when correctness, explanation, traceability, or human comprehension is required.

8 The Self-Defeat of Philosophical Objection

Philosophical objections to AI are often valuable. They discipline technological enthusiasm and force institutions to ask what they are doing. However, philosophy is not a sovereign tribunal outside institutional life. It is itself an institutional practice.

Philosophy can become self-defeating in at least three ways.

First, philosophy may object that AI lacks consciousness while ignoring that many institutional processes already operate through non-conscious symbolic systems: forms, rules, precedents, grading rubrics, templates, bureaucratic classifications, and databases. If lack of consciousness disqualifies AI from institutional assistance, then much of existing institutional machinery becomes suspect.

Second, philosophy may criticize AI for opacity while tolerating opacity in philosophical gatekeeping. Academic hierarchies, canon formation, peer review, jargon, citation networks, and departmental authority can themselves be opaque.

Third, philosophy may demand accountability from AI systems while refusing accountability for its own institutional costs: exclusion, delay, interpretive monopoly, credentialism, and insulation from empirical audit.

Proposition 4 (Reciprocity of critique). *A philosophical critique of AI is institutionally legitimate only if the critic accepts equivalent scrutiny of the philosophical institution producing the critique.*

Proof. A critique that applies a standard to another object while exempting itself from that standard is asymmetrical. If the critique concerns opacity, power, accountability, or legitimacy, then the institutional position of the critic is relevant to the validity of the critique as institutional guidance. Therefore a philosophical critique of AI has institutional legitimacy only when philosophy accepts reciprocal scrutiny. $\square$

The point is not that philosophy is useless. The point is that philosophy must remain answerable to its own demands. AI does not defeat philosophy. AI reveals whether philosophy is willing to be examined by its own standards.

9 Philosophy as Its Own Anathema

The deeper problem is not merely self-defeat. Philosophy may become its own anathema. That is, it may become the very thing it condemns.

If philosophy condemns dogmatism, it may become dogmatic. If it condemns unaccountable authority, it may become an unaccountable authority. If it condemns mechanistic thinking, it may become a procedural machine of gatekeeping. If it condemns institutional power, it may protect its own institutional power. If it condemns empty language, it may become empty language protected by professional prestige.

Definition 5 (Philosophical anathema). *Philosophical anathema occurs when philosophy reproduces, within its own institutional practice, the very defect it condemns in its object of critique.*

This idea matters for AI because philosophical resistance to AI often takes the form of categorical prohibition. But categorical prohibition can become a way of protecting existing authority over represented thought. If AI makes archives more searchable, arguments more comparable, and texts more auditable, then institutions built on control over interpretation may experience AI as a threat.

Theorem 1 (AI as reciprocal audit). *AI becomes institutionally disruptive not only when it automates tasks, but when it exposes whether institutions can tolerate the auditability of their own represented thought.*

Proof. Institutions preserve authority partly through control over represented thought: archives, doctrines, procedures, classifications, and interpretations. AI lowers the cost of searching, comparing, summarizing, and generating from such representations. Therefore it lowers the cost of auditing institutional claims. If an institution depends on opacity or interpretive monopoly, AI threatens its authority. If an institution is willing to be audited, AI strengthens its capacity. Hence AI is disruptive not merely as automation but as reciprocal audit. $\square$

10 Accountable Integration

The correct response is neither uncritical adoption nor philosophical rejection. Institutions should integrate AI under rules that preserve accountability, transparency, human responsibility, and auditability.

A practical framework can be stated as follows.

Principle	Institutional requirement
Human responsibility	A named human remains accountable for institutional output.
Corpus awareness	Users know what body of material AI is drawing from.
Traceability	Important claims should be traceable to sources or evidence.
Role distinction	Drafting, advising, deciding, and authorizing are separated.
Error discipline	AI output is checked before institutional reliance.
Reciprocal audit	Critics of AI accept scrutiny of their own procedures.

10.1 Algorithm for Institutional AI Use

Algorithm: Accountable AI Integration

1. Identify the institutional corpus D .
2. Classify the task as query, organization, understanding, analysis, summarization, or generation.
3. Determine the risk level of the task.
4. If the task affects rights, grades, employment, legal status, public safety, or major resource allocation, require human review.

5. Use AI to produce an output O_A .
6. Require a responsible human agent to verify O_A .
7. Record the role of AI: assistant, summarizer, drafter, analyst, or recommender.
8. Preserve evidence, citations, or reasoning where relevant.
9. Permit institutional appeal, correction, or audit.
10. Periodically evaluate whether AI use improves accuracy, fairness, speed, and legitimacy.

11 A Diagram of the Argument

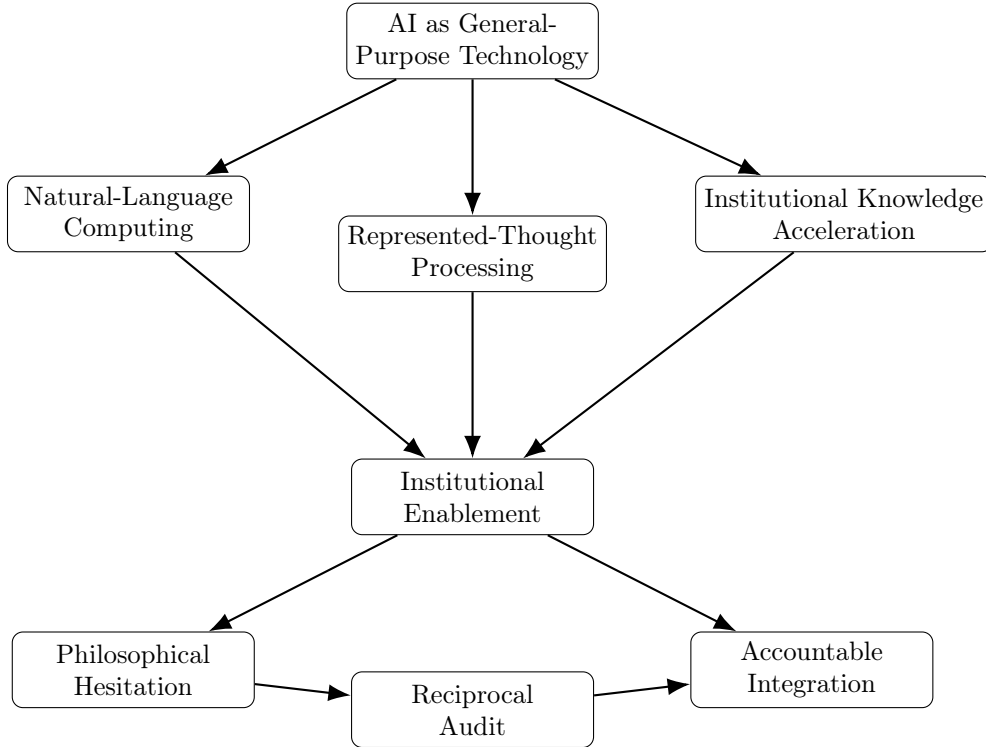


Figure 1: AI as a general-purpose technology producing institutional enablement and philosophical contestation.

12 Implications

The framework has several implications.

First, AI policy should not be built on the false question of whether AI is a human-like mind. For most institutional purposes, the relevant question is whether AI can responsibly process represented thought under accountable human supervision.

Second, institutions should distinguish between AI as tool, AI as assistant, AI as recommender, and AI as decision system. The first three may be widely appropriate. The fourth requires much stricter governance.

Third, philosophical critique should be preserved but not absolutized. Philosophy helps institutions clarify concepts, but it cannot claim immunity from institutional analysis.

Fourth, AI adoption should be evaluated empirically. Institutions should ask whether AI improves accuracy, speed, accessibility, fairness, transparency, and accountability. Philosophical concerns should inform these evaluations, not replace them.

Fifth, AI may democratize access to institutional knowledge. If ordinary users can query complex corpora through natural language, then expertise becomes more accessible. But this democratization is incomplete unless institutions design systems that preserve evidence, context, and contestability.

13 Conclusion

Artificial intelligence is not an institution, nor is it a human-like mind. It is a general-purpose technology that changes how individuals interact with computers. Its distinctive contribution is the transformation of represented thought into a more queryable, organizable, analyzable, summarizable, and generative medium.

Institutions hesitate to adopt AI because AI unsettles philosophical assumptions about agency, authorship, authenticity, accountability, judgment, and knowledge. These concerns are serious. But philosophy is not outside the institutional world. It can become self-defeating when it exempts itself from the standards it applies to AI. It becomes its own anathema when it reproduces the dogmatism, opacity, exclusion, or authority it condemns.

The institutional future of AI should therefore be neither blind technological adoption nor philosophical prohibition. It should be accountable integration. AI should be used where it improves human capacity to work with represented thought, but under rules that preserve responsibility, traceability, review, and reciprocal audit.

The central question is not whether AI defeats philosophy. The central question is whether philosophy, and every other institution of represented thought, can survive being examined by the tools it seeks to judge.

References

- [1] T. Bresnahan and M. Trajtenberg. General purpose technologies: Engines of growth? *Journal of Econometrics*, 65(1):83–108, 1995.
- [2] P. A. David. The dynamo and the computer: An historical perspective on the modern productivity paradox. *American Economic Review*, 80(2):355–361, 1990.
- [3] E. Brynjolfsson and A. McAfee. *The Second Machine Age*. W. W. Norton, 2014.
- [4] D. C. North. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press, 1990.
- [5] H. A. Simon. *Administrative Behavior*. Macmillan, 1947.
- [6] J. R. Searle. Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3):417–457, 1980.
- [7] A. Giddens. *The Constitution of Society*. University of California Press, 1984.
- [8] S. Ghosh. *The Self-Defeat of Philosophy*. Kolkata, India, 2026.
- [9] S. Ghosh. *Philosophy as Its Own Anathema*. Kolkata, India, 2026.

Glossary

Artificial intelligence

Computational systems capable of performing tasks such as language generation, classification, prediction, summarization, and analysis.

General-purpose technology

A technology with broad applicability, continuing improvement, and complementarities across many sectors.

Natural-language computing

The use of ordinary language as an interface for computational operation.

Represented thought

Human or institutional cognition externalized into symbolic form, including documents, laws, code, papers, transcripts, and records.

Institutional knowledge acceleration

The reduction in time, cost, or cognitive burden required to query, organize, understand, analyze, summarize, or generate institutional knowledge.

Philosophical self-defeat

The failure of philosophy to submit itself to the standards it applies to other objects of critique.

Philosophical anathema

The condition in which philosophy becomes the very defect it condemns.

Reciprocal audit

A principle requiring that institutions criticizing AI accept scrutiny of their own procedures, assumptions, incentives, and authority.

Accountable integration

The institutional use of AI under human responsibility, traceability, review, and correction.

The End